ASSIGNMENT: Bank Marketing

Name: Nandini Soni

Problem Statement

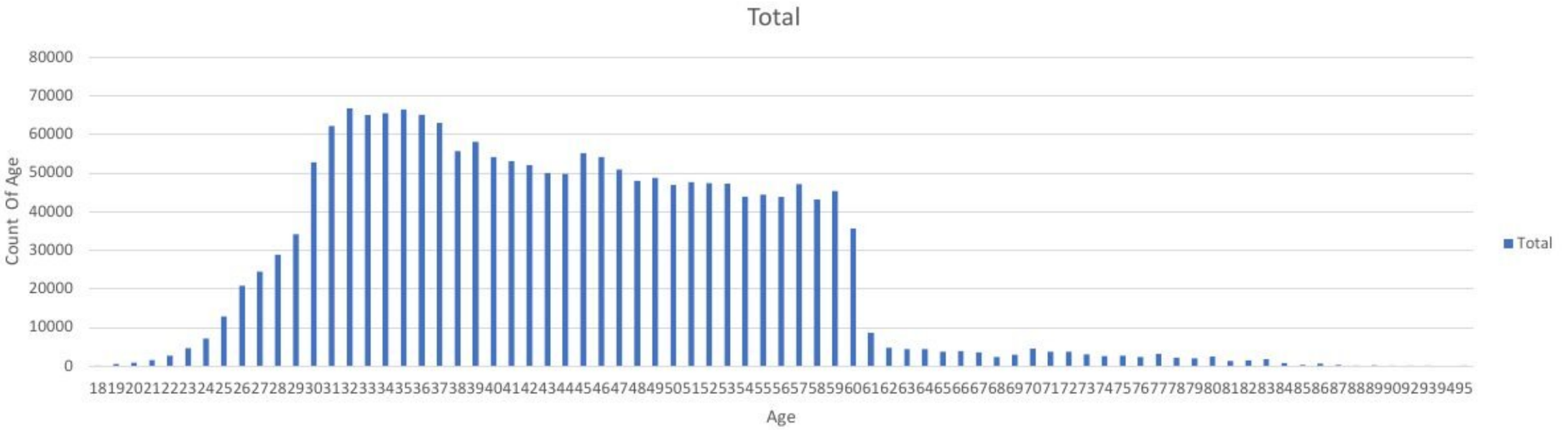
Bank of Corporate conducted a telemarketing campaign for one of its financial products, 'Term deposits', to build a long-term relationship with the existing customers. Your goal is to identify the target customers for the term deposits from the pool of the bank's existing customers. You should also capture the key driving factors (or driver variables) behind the successful conversion of a customer, i.e., the customer opening a term deposit account with the bank.

Assignment Objective

To identify the target customers and the driving factors behind a successful customer conversion for future marketing campaigns.

PART I : Univariate Analysis

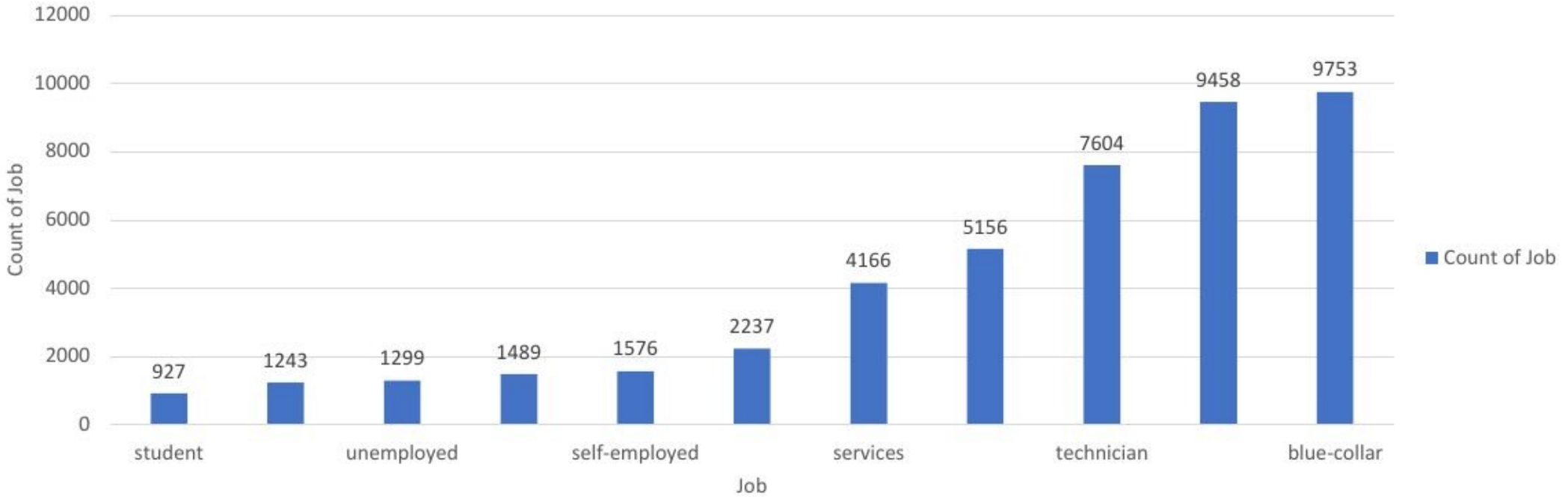
by Age



Observation: Target marketing campaigns towards the most responsive age groups

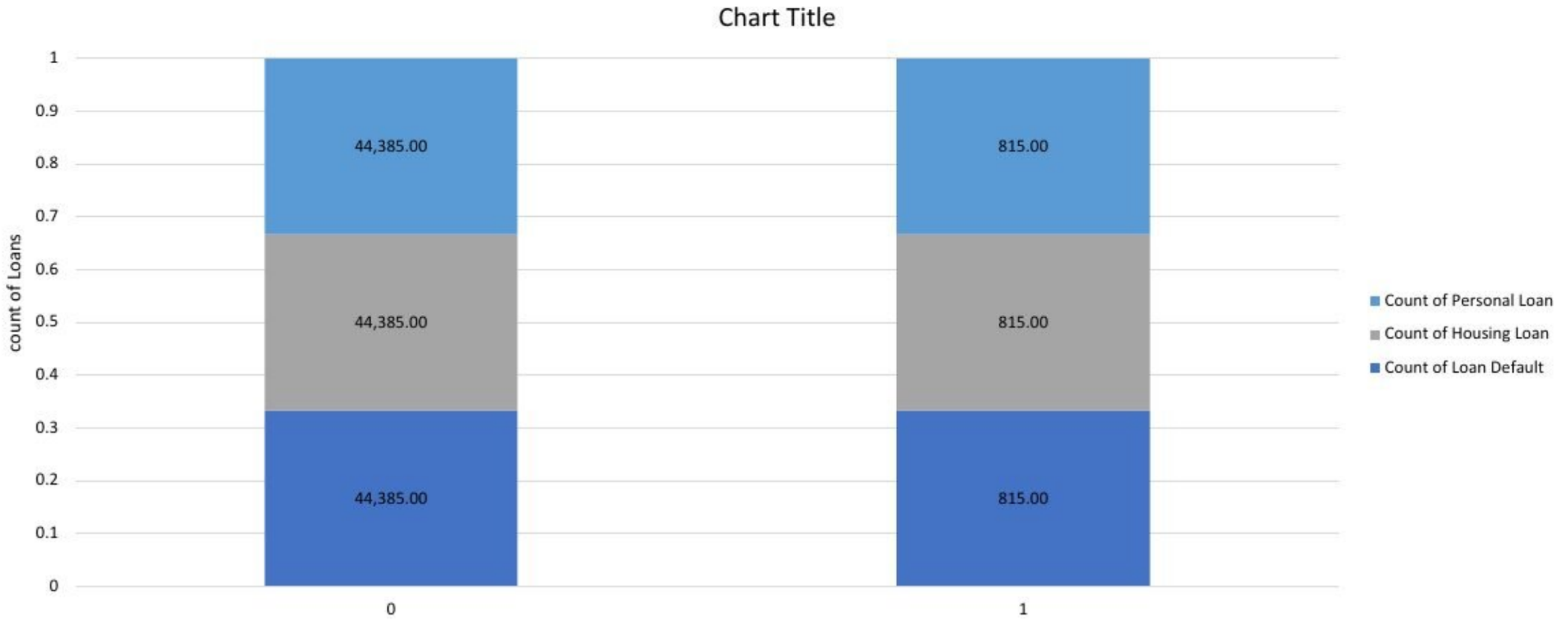
PART I : Univariate Analysis

By Job



Observation: Customize product offerings for dominant job categories ‘blue-collar’ job with highest count of job

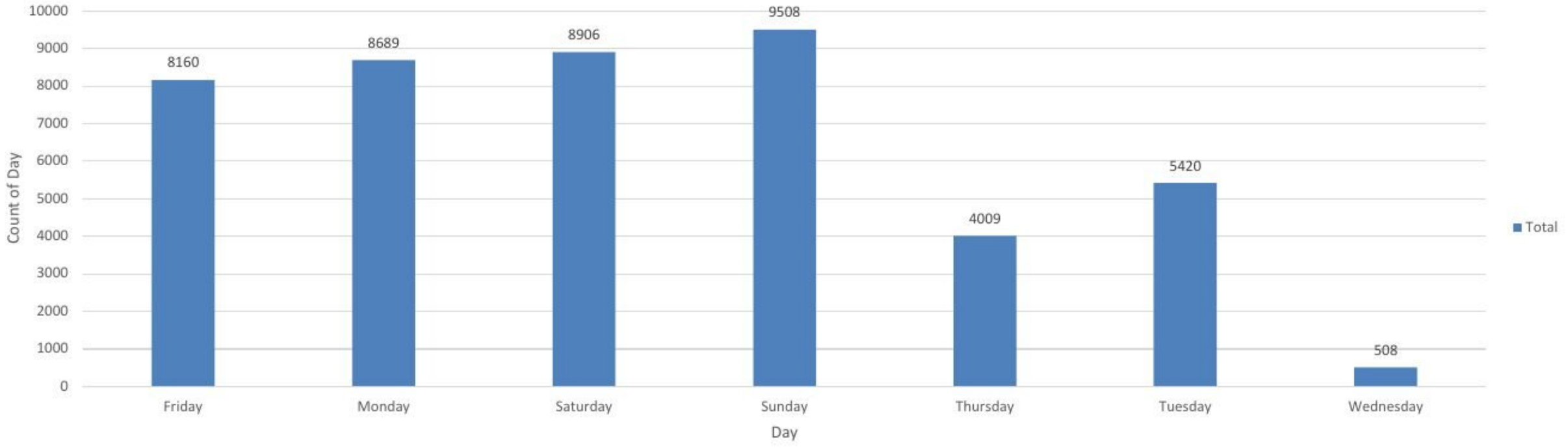
PART I : Univariate Analysis
by loans



Observation: Developed specialized loan products and risk mitigation strategies for prevalent loan types and high-risk segments.

PART I : Univariate Analysis

by Day



Observation: Increase marketing efforts on days with ‘Sundays’ on higher customer engagement.

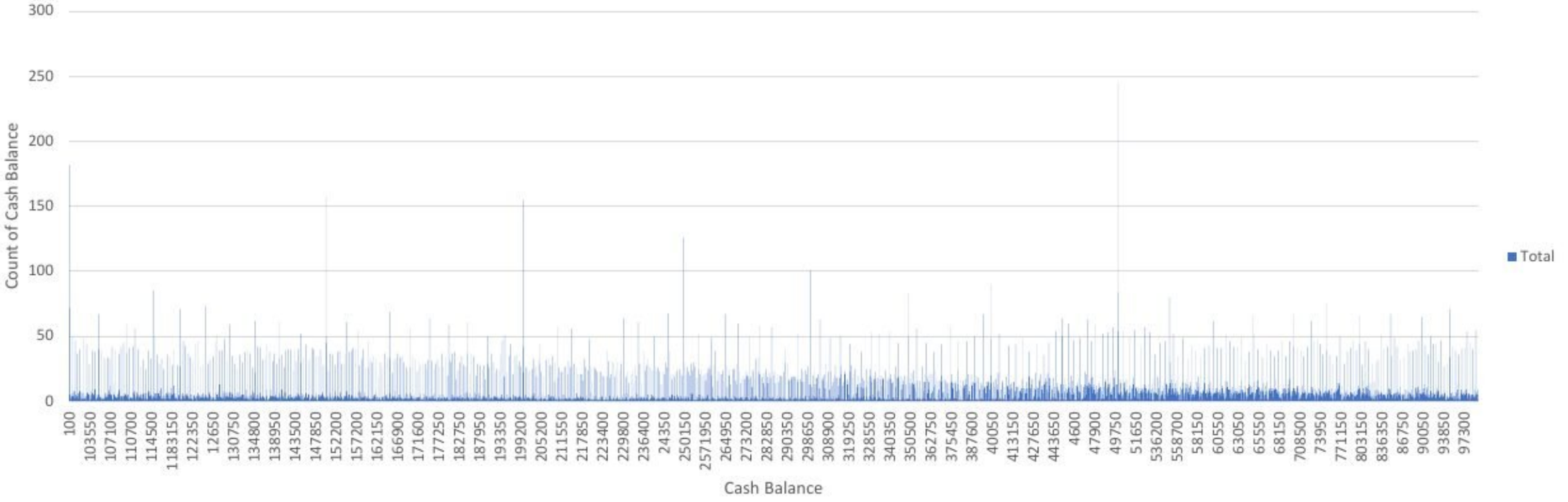
PART I : Univariate Analysis
by Month



Observation: Align marketing strategies with ’May’ on peak months for better results.

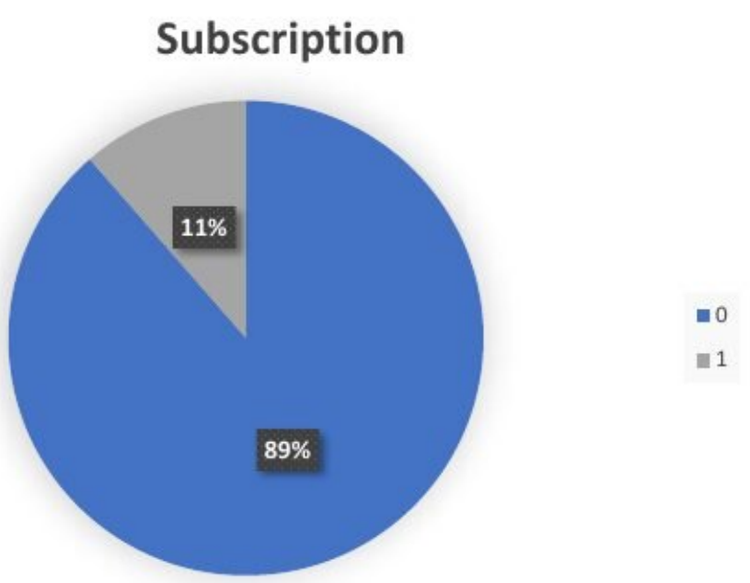
PART I : Univariate Analysis

by Cash Balance

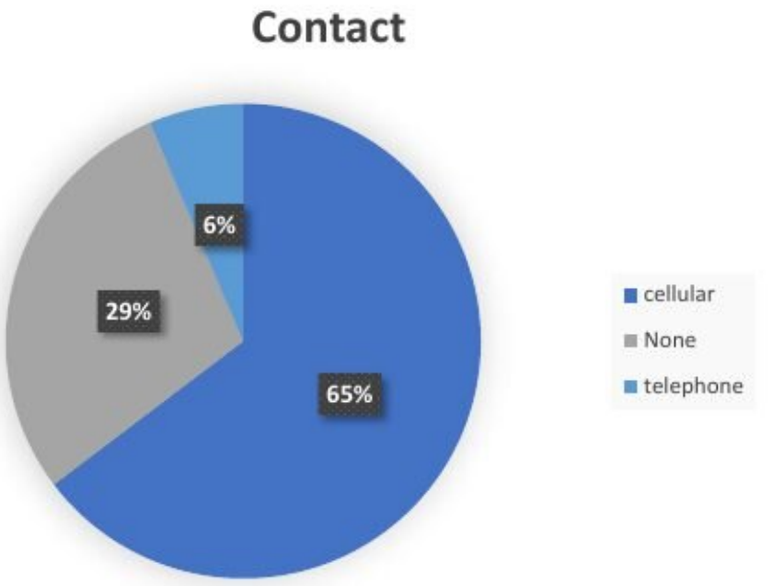


Observation: Offer tiered services based on customer balance segments

Segmented Univariate Analysis

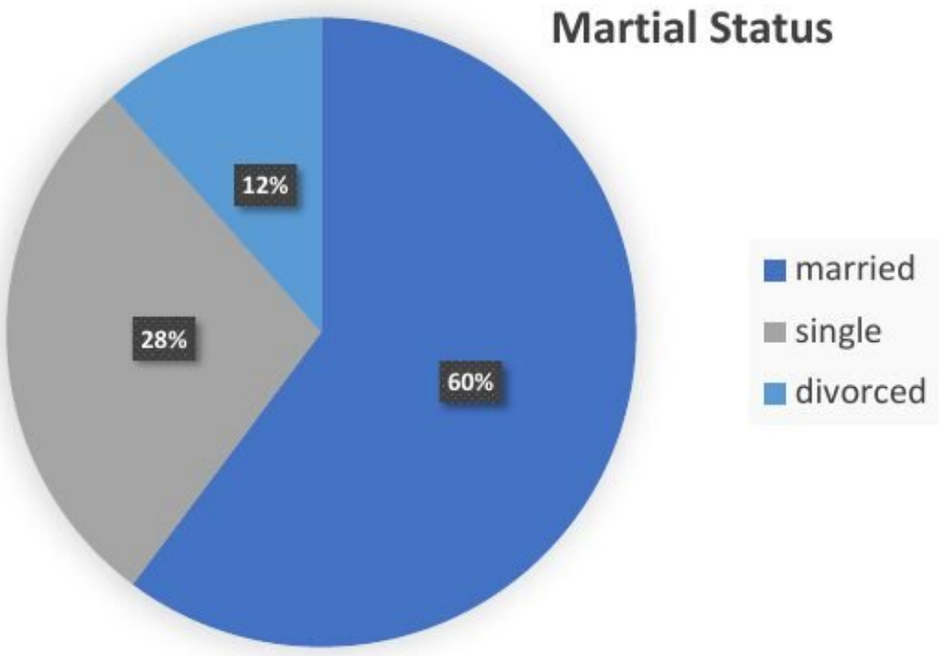


Observation: Focus subscription offers with 89% higher rates of acceptance

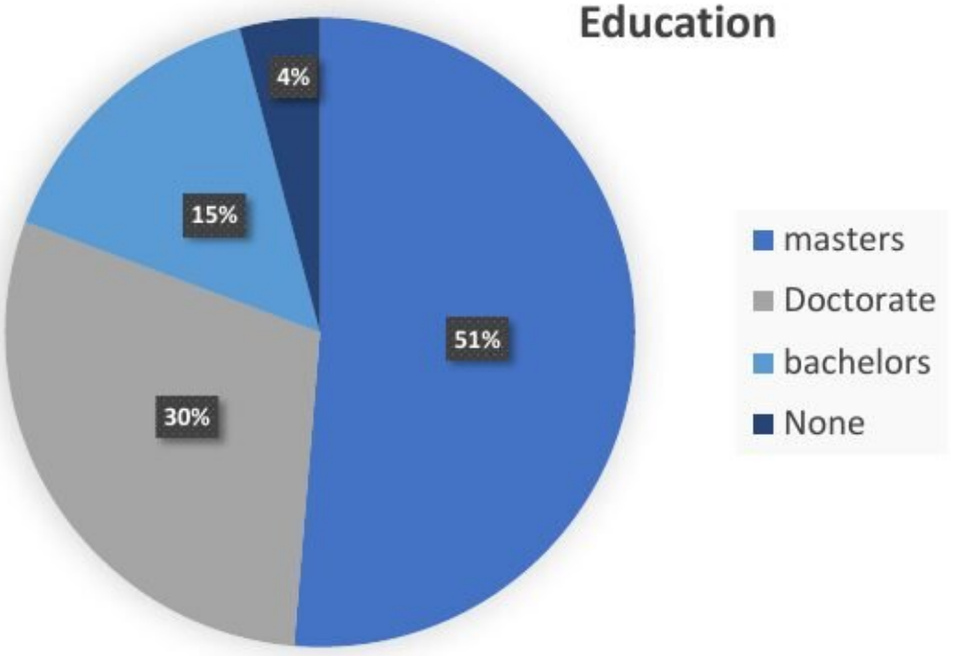


Observation: Segment the preferred contact methods for 65% of cellular users

Segmented Univariate Analysis

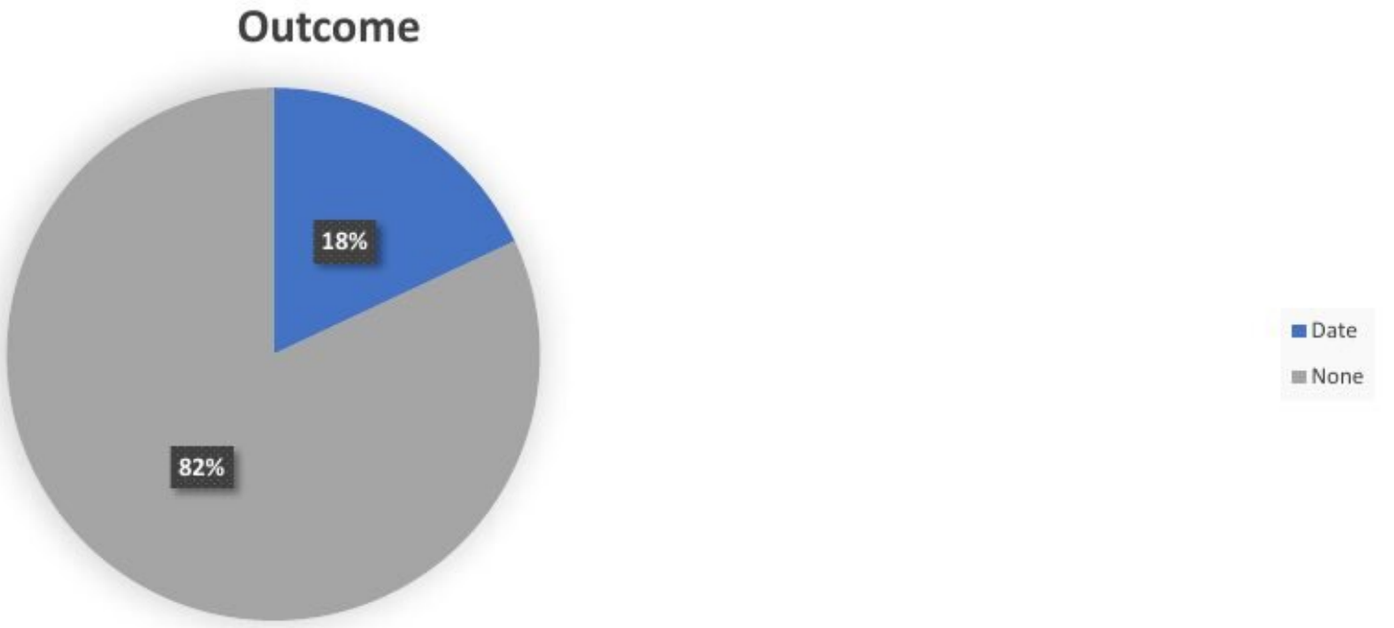


Observation: Assess the distribution of marital status of 60% to understand the demographic composition of single



Observation: Segment the preferred masters by 51% to see which methods are most effective for each education segment

Segmented Univariate Analysis



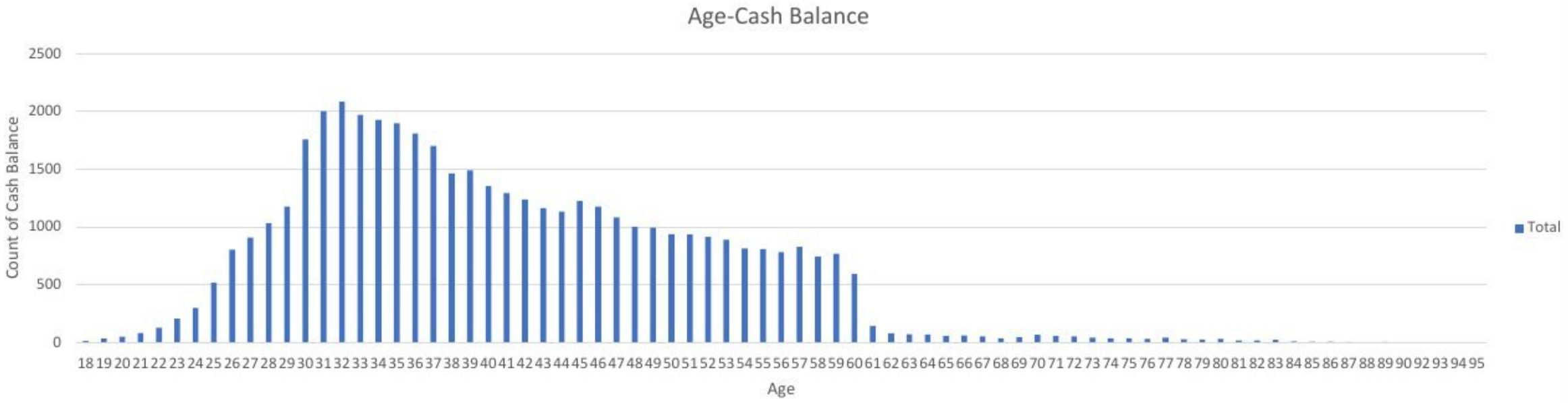
Observation: Determine the preferred Octcome(oputcome) for different age groups

Data Cleaning

SEGMENTED UNIVARIATE ANALYSIS

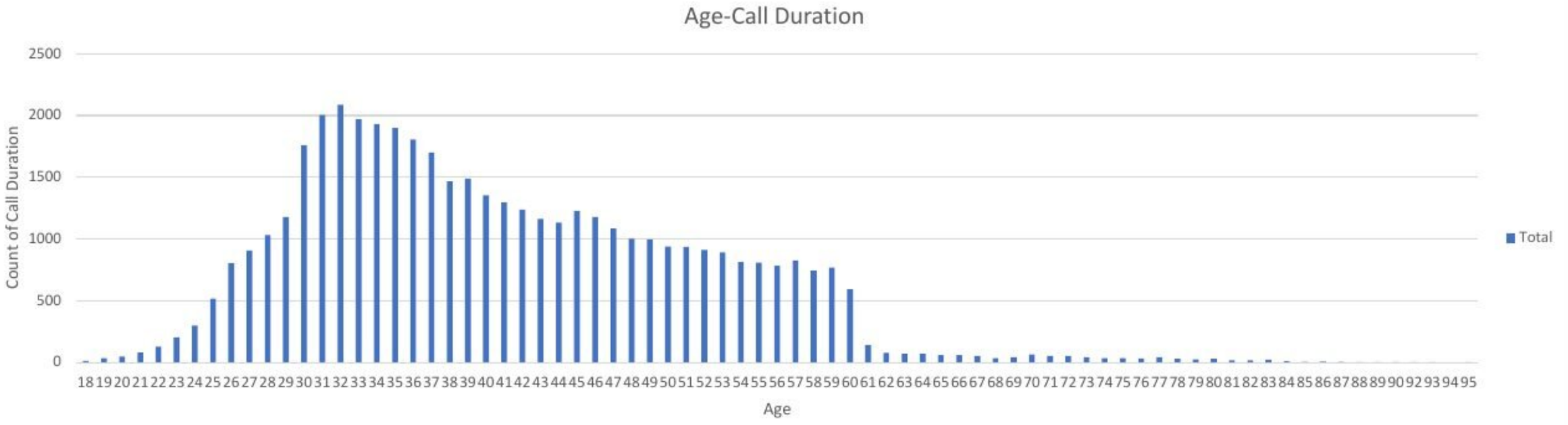
- Remove duplicate rows
- Remove rows with Blank cells (Loan default, subscription, poutcome)
- Data conversions: In Age, negative to positive
- In Job, convert admin. admin
- In Loan Default, Housing Low: Personal Loan, and Subscription, replace 'yes' with 1 and 'No' with 0
- Convert the Date from text to Date format
- Create the new columns Day and Month from the Date column
- In Call Duration, convert Hrs, Mins, Sec to seconds
- Remove outlier in 'Previous' column

PART II : Bivariate Analysis



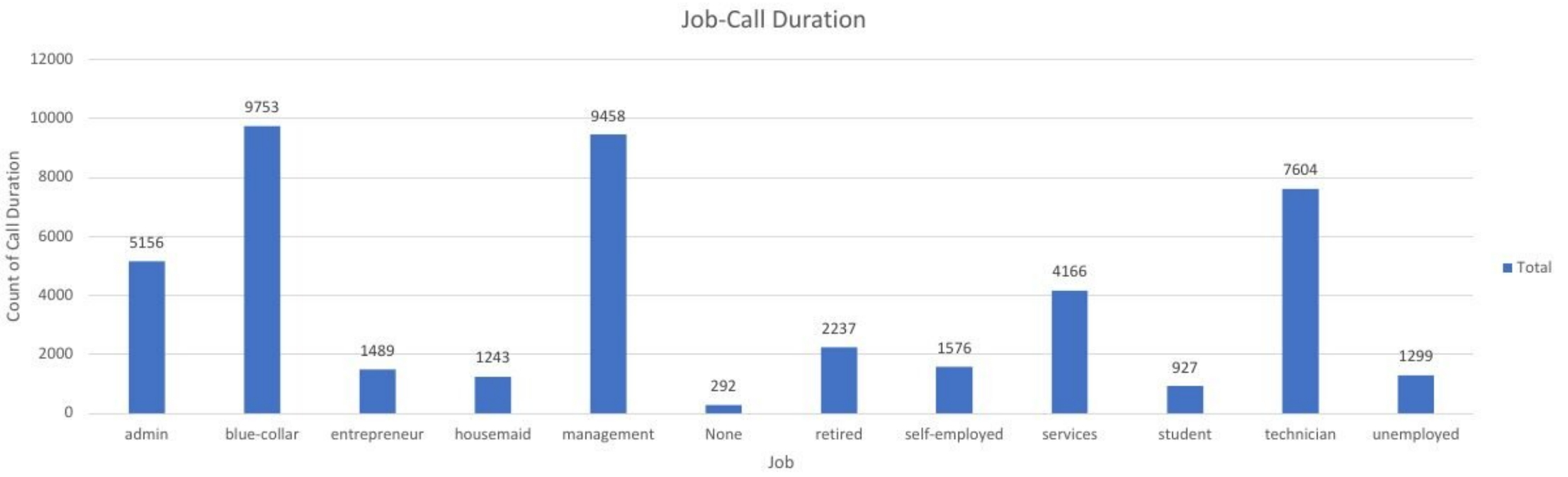
Observation: Examine the Cash Balance distribution among different age groups

PART II : Bivariate Analysis



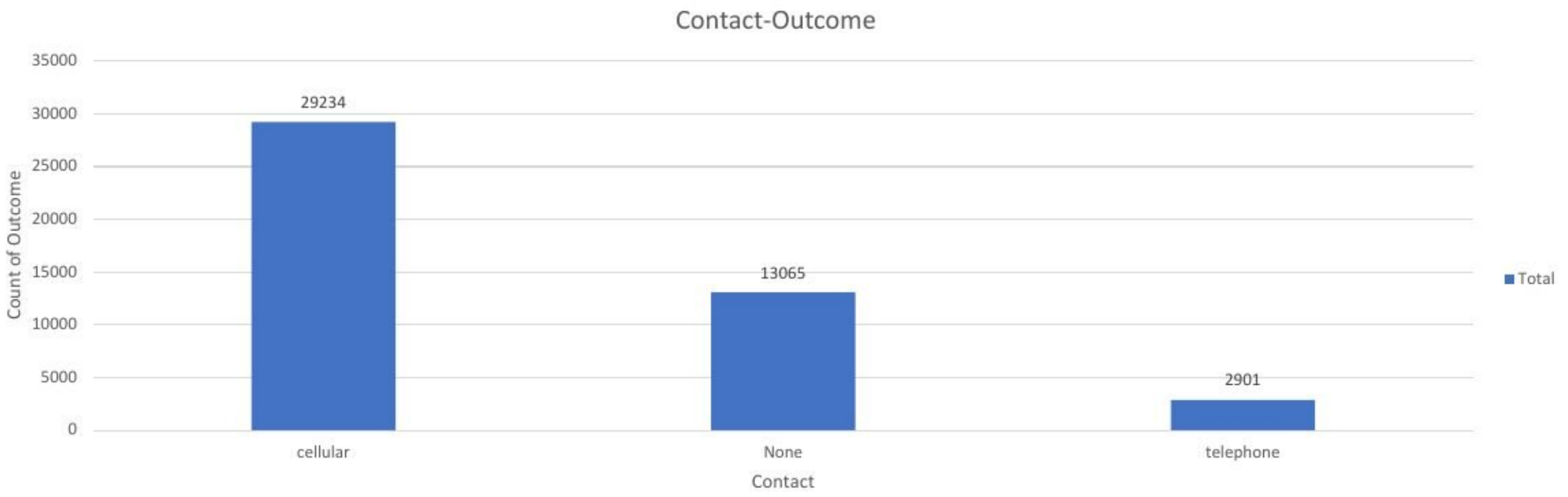
Observation: Examine the CallDuration distribution among different age groups

PART II : Bivariate Analysis



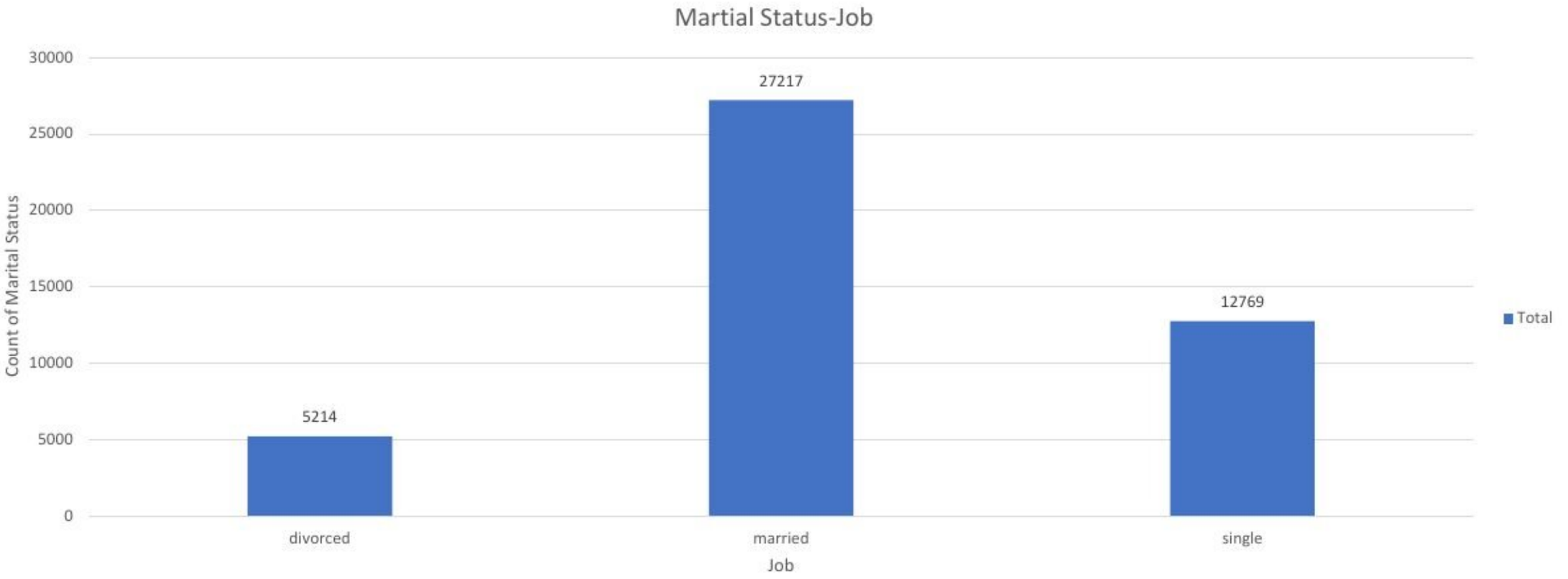
Observation: Analyze which contact methods are preferred by different job types to tailor communication strategies.

PART II : Bivariate Analysis



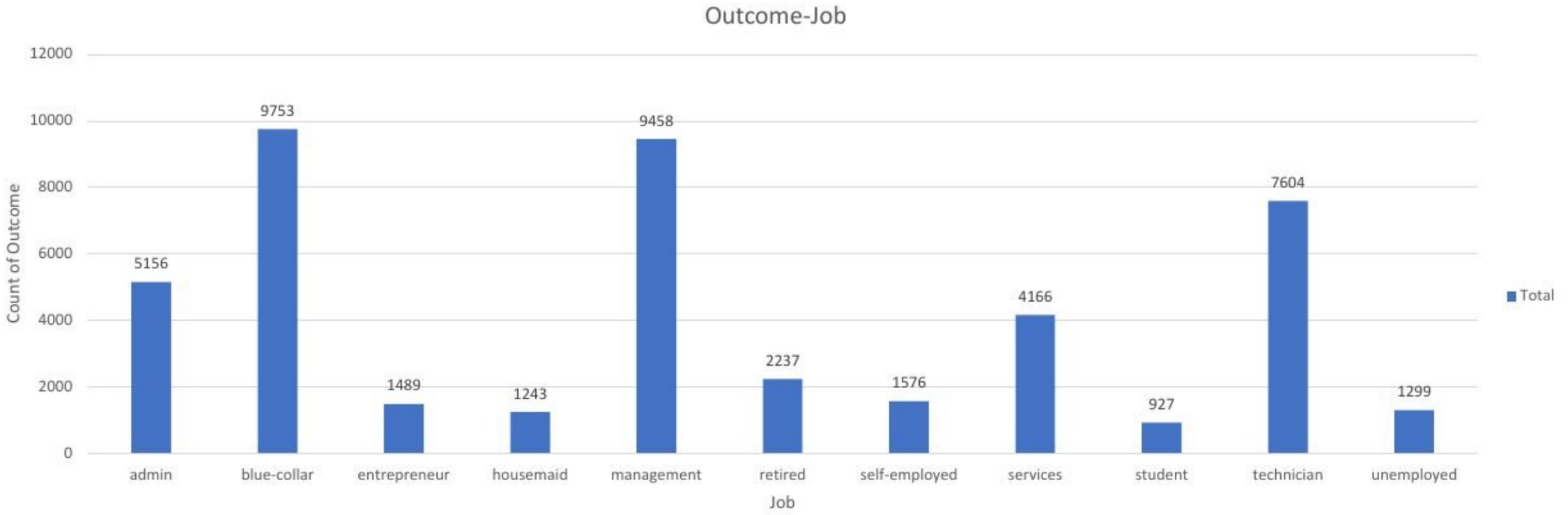
Observation: Evaluate the effectiveness of each contact method based on the number of contacts made, to understand which method yields the highest response rates

PART II : Bivariate Analysis



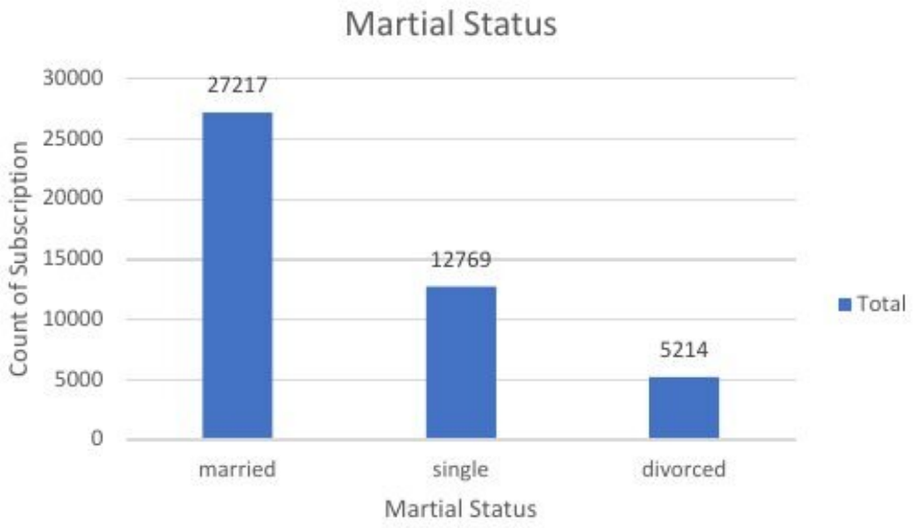
Observation: Assess the marital status distribution within different job categories

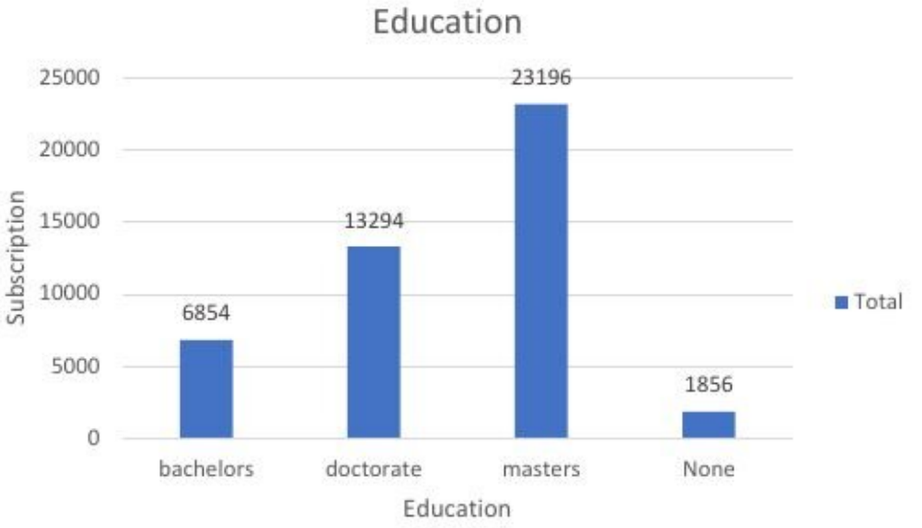
PART II : Bivariate Analysis



Observation: Analyze the correlation between outcome of different job types

PART II : Bivariate Analysis





Observation: Examine the distribution of customers' marital statuses on subscription with married martial status be the highest

Observation: Evaluate the educational levels of customers on subscriptions with masters education level be the highest

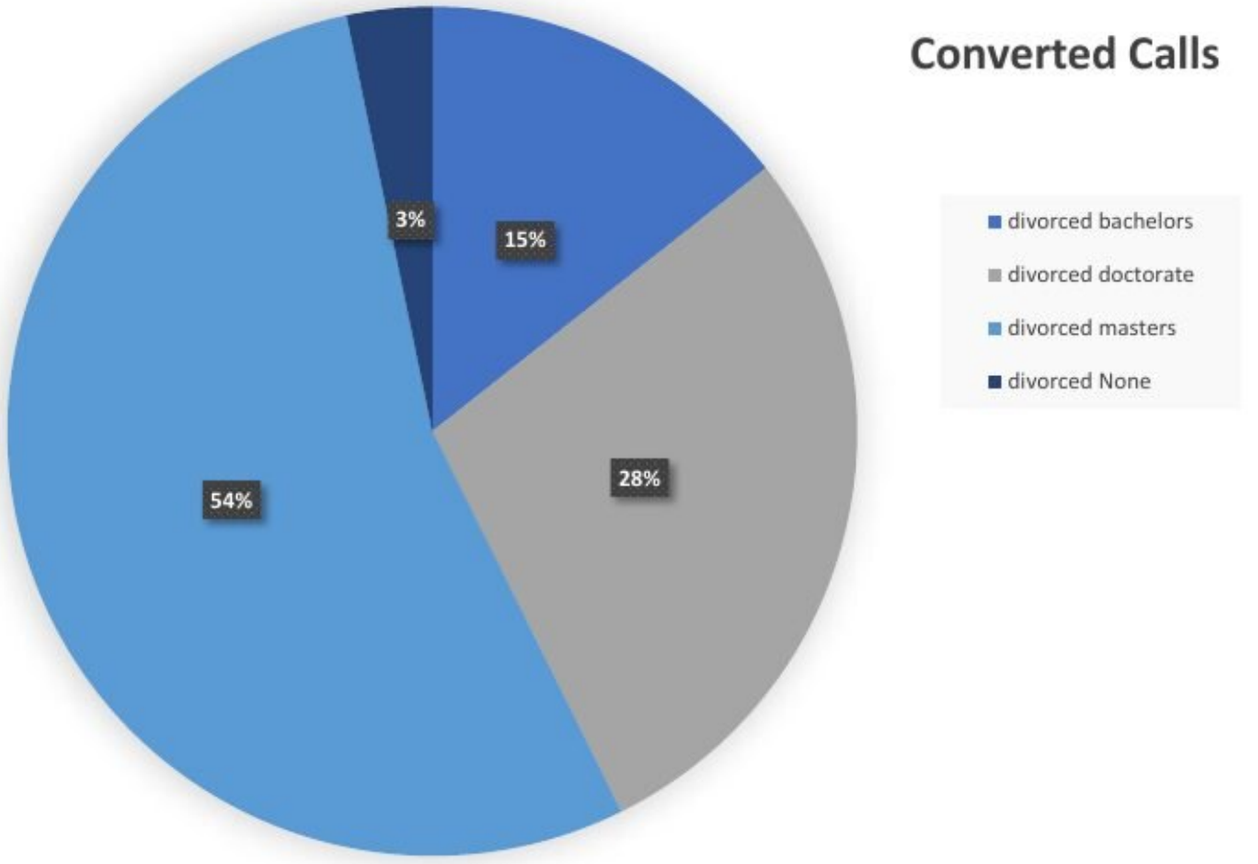
Data Cleaning

SEGMENTED BIVARIATE ANALYSIS

- **Identify and Handle Missing Values**: Check for and impute or delete missing data.
- **Remove Duplicates:** Use tools to detect and eliminate exact and contextual duplicates.
- **Address Outliers:** Use visual and statistical methods to identify and decide whether to retain, remove, or transform outliers.
- **Ensure Consistent Formatting:** Verify correct data types, consistent units, and standardized categorical labels.
- **Correct Spelling Inconsistencies:** Standardize spelling errors in categorical variables.
- **Create Derived Variables:** Form new variables by combining or transforming existing ones.
- **Validate Data:** Cross-check with the original data and perform peer reviews for accuracy.

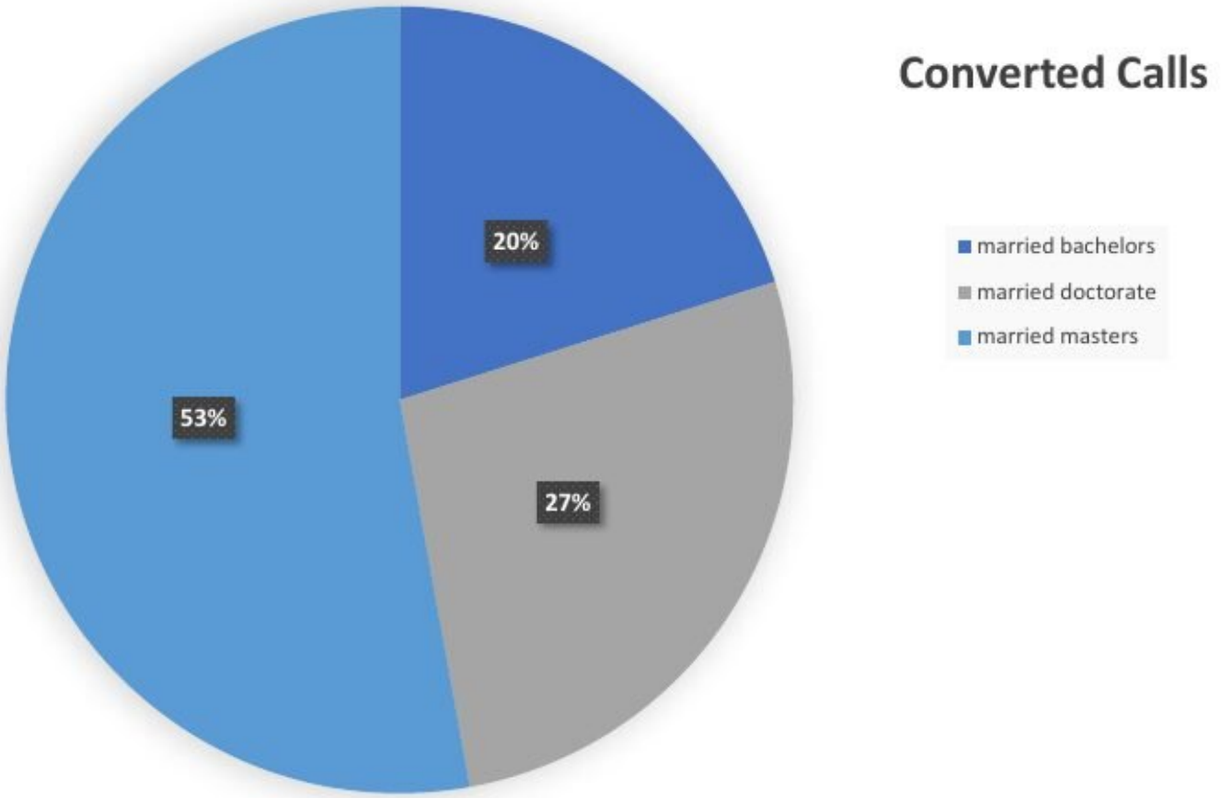
Analysis Result

Divorced Martial Status Segmented by Education



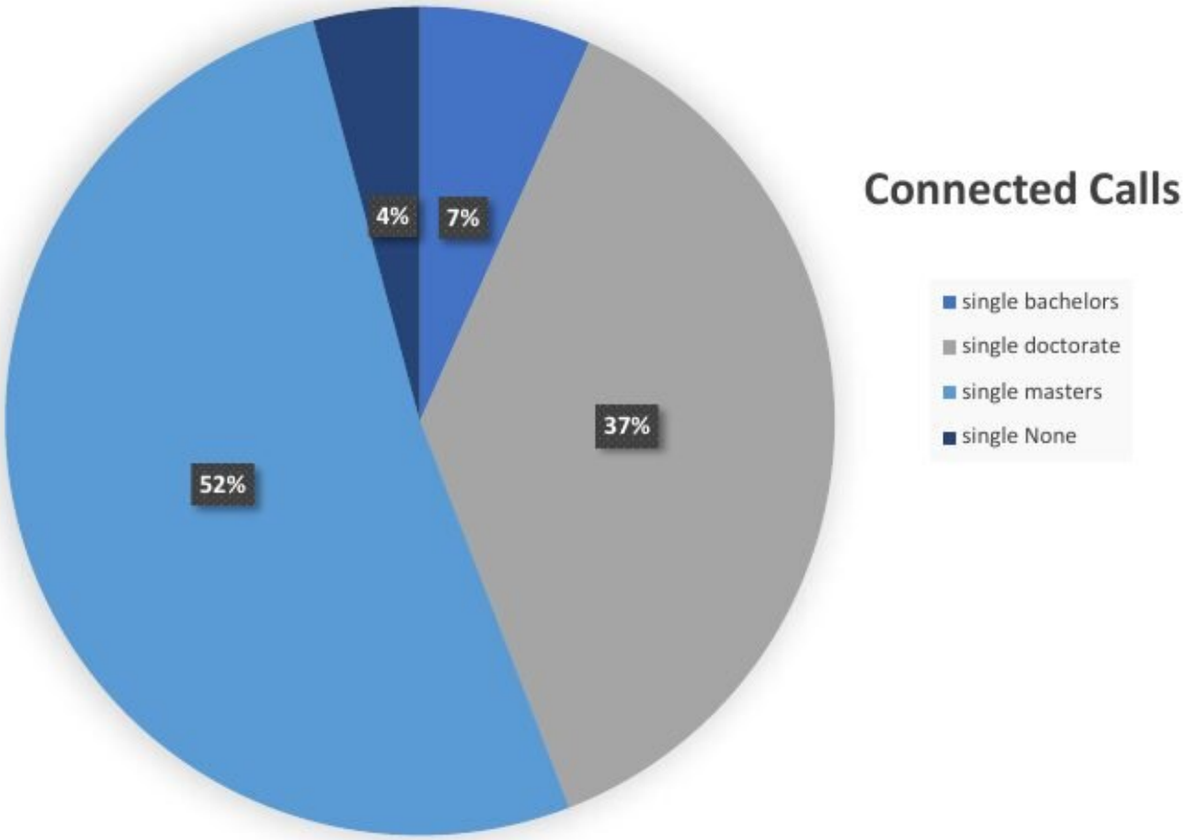
Analysis Result

Married Martial Status Segmented by Education



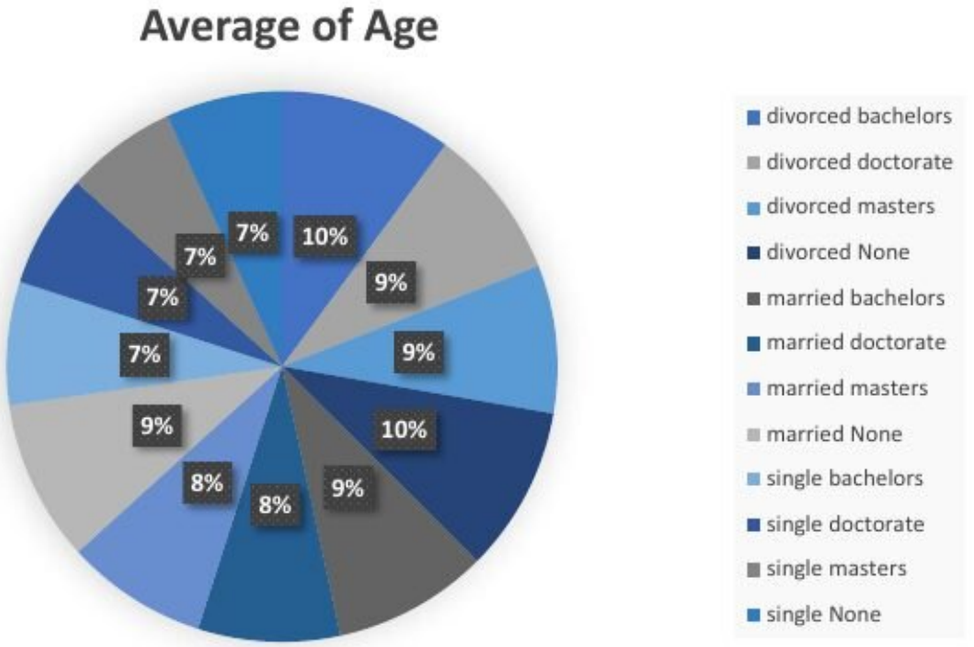
Analysis Result

Single Martial Status Segmented by Education



Analysis Result

Average age of martial Status Segmented by Education



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PART III: Major Insights

Insight 1: Age :

The majority of customers fall within the 30-50 age group

Insight 2: Job :

The most common job categories are 'blue-collar' and 'management

Insight 3: Marital Status :

Most customers are married

Insight 4: Cash Balance :

The majority of customers have cash balances in the range of 1000-5000

Insight 5: Loan Status :

A significant proportion of customers have housing loans, and a smaller percentage have personal loans